

# Linear Algebra

## Solutions to Practice Problems

### Chapter 1 — Matrices

#### How This Pairs With Your Notes

This is the companion answer key to *LinearAlgebra\_Ch1\_Matrices\_Notes.pdf*. Each problem is restated in an orange box (so this file is self-contained – you don't need the notes open side by side), followed by a full worked solution in teal. Try each problem yourself first; the value of a practice problem comes from attempting it before you check the answer.

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## 1 Solutions – §1.1 (Definition of a Matrix)

### Problem 1.1.1

Write down the order of  $A = \begin{pmatrix} 2 & 5 & 1 \\ 0 & -1 & 4 \end{pmatrix}$  and identify  $a_{11}$ ,  $a_{23}$ .

### Solution

$A$  has 2 rows and 3 columns, so its order is  $2 \times 3$ . Reading off entries (row, column):  $a_{11} = 2$  (row 1, column 1) and  $a_{23} = 4$  (row 2, column 3).

### Problem 1.1.2

Construct the augmented matrix for the system  $2x - y + z = 4$ ,  $x + 3y - z = 0$ .

### Solution

Read the coefficients of  $x, y, z$  off each equation, in order, then the right-hand side after the divider:

$$\left[ \begin{array}{ccc|c} 2 & -1 & 1 & 4 \\ 1 & 3 & -1 & 0 \end{array} \right].$$

### Problem 1.1.3

Give a  $3 \times 3$  diagonal matrix that is *not* a scalar matrix, then one that *is* a scalar matrix.

**Solution**

Not scalar (diagonal entries differ):  $\text{diag}(1, 2, 3) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ . Scalar (all diagonal entries equal, here to 5):  $5I_3 = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$ . Any multiple of  $I_3$  works for the second part – the defining feature of a scalar matrix is that *every* diagonal entry is the same number.

**Problem 1.1.4**

Is  $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$  upper triangular, lower triangular, both, or neither? What about  $\begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix}$ ?

**Solution**

For  $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$ : every entry below the diagonal ( $a_{21}, a_{31}, a_{32}$ ) is 0, so it *is* upper triangular. But  $a_{12} = 2 \neq 0$  (above the diagonal), so it is **not** lower triangular. Answer: **upper triangular only**.

For  $\begin{pmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{pmatrix} = 7I_3$ : every off-diagonal entry is 0, so all entries above *and* below the diagonal vanish. Answer: **both** upper and lower triangular (as is true of every diagonal matrix).

**Problem 1.1.5**

Write out  $I_4$  in full.

**Solution**

$$I_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

**Problem 1.1.6 (Challenge)**

Can a non-square matrix ever be called “diagonal” under Definition 1.1.5? Explain why the definition only makes sense for square matrices.

**Solution**

Strictly by the wording “ $a_{ij} = 0$  for  $i \neq j$ ”, you *could* technically apply it to a non-square matrix – e.g. a  $2 \times 3$  matrix with  $a_{11}, a_{22}$  possibly nonzero and everything else 0. But this isn’t useful, for two reasons. First, the “principal diagonal” entries  $a_{11}, \dots, a_{nn}$  are only defined up to  $n = \min(m, n)$  when  $m \neq n$ , so some rows or columns are left with no diagonal entry at all – the concept doesn’t fully cover the matrix. Second, and more importantly, the whole *point* of a diagonal matrix is to represent independent scaling along each coordinate axis of a space mapped to *itself* ( $D : \mathbb{R}^n \rightarrow \mathbb{R}^n$ ); that only makes sense when domain and codomain have the same dimension, i.e. when the matrix is square. So the definition is conventionally restricted to square matrices.

## 2 Solutions – §1.2 (Operations on Matrices)

### Problem 1.2.1

For  $A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \\ -1 & 4 \end{pmatrix}$ , find  $A^t$  and state its order.

### Solution

$A$  has order  $3 \times 2$ , so  $A^t$  has order  $2 \times 3$ . Reading the rows of  $A$  off as the columns of  $A^t$ :

$$A^t = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 4 \end{pmatrix}.$$

### Problem 1.2.2

Let  $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ ,  $B = \begin{pmatrix} 1 & 4 \\ 2 & 0 \end{pmatrix}$ . Compute  $AB$  and  $BA$ . Do they commute?

### Solution

$$AB = \begin{pmatrix} 2(1) + 1(2) & 2(4) + 1(0) \\ 0(1) + 3(2) & 0(4) + 3(0) \end{pmatrix} = \begin{pmatrix} 4 & 8 \\ 6 & 0 \end{pmatrix}.$$

$$BA = \begin{pmatrix} 1(2) + 4(0) & 1(1) + 4(3) \\ 2(2) + 0(0) & 2(1) + 0(3) \end{pmatrix} = \begin{pmatrix} 2 & 13 \\ 4 & 2 \end{pmatrix}.$$

Since  $AB \neq BA$  (they don't even share a single entry),  $A$  and  $B$  **do not commute**.

### Problem 1.2.3

Let  $A$  be  $2 \times 3$  and  $B$  be  $3 \times 5$ . Explain, using Definition 1.2.8, why  $AB$  is defined but  $BA$  is not.

### Solution

$AB$  needs ( $\#$  columns of  $A$ ) = ( $\#$  rows of  $B$ ): here  $3 = 3$ , so  $AB$  is defined and has order  $2 \times 5$ .  $BA$  needs ( $\#$  columns of  $B$ ) = ( $\#$  rows of  $A$ ): here that would require  $5 = 2$ , which is false. So  $BA$  is **not defined** – there's no way to pair up a row of  $B$  (length 5) with a column of  $A$  (length 2) to form a dot product.

### Problem 1.2.4

Compute  $DA$  and  $AD$  for  $D = \text{diag}(2, -1)$  and  $A = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix}$ , and check your answer against Theorem 1.2.11(5).

### Solution

By direct multiplication (or by the shortcut in Theorem 1.2.11(5)):

$$DA = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 6 & 8 \\ -5 & -6 \end{pmatrix}, \quad AD = \begin{pmatrix} 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 6 & -4 \\ 10 & -6 \end{pmatrix}.$$

Check against the shortcut: row 1 of  $A$  is  $(3, 4)$ , scaled by  $d_1 = 2$  gives  $(6, 8)$  – matches row 1 of  $DA$ . Row 2 of  $A$  is  $(5, 6)$ , scaled by  $d_2 = -1$  gives  $(-5, -6)$  – matches row 2 of  $DA$ . ✓ Similarly, column 1 of  $A$  is  $(3, 5)^t$ , scaled by  $d_1 = 2$  gives  $(6, 10)^t$  – matches column 1 of  $AD$ ; column 2 of  $A$  is  $(4, 6)^t$ , scaled by  $d_2 = -1$  gives  $(-4, -6)^t$  – matches column 2 of  $AD$ . ✓

**Problem 1.2.5**

If  $A + B = A$ , show (directly from the definitions) that  $B = 0$ .

**Solution**

Write  $A = [a_{ij}]$ ,  $B = [b_{ij}]$ . The equation  $A + B = A$  means, entrywise,  $a_{ij} + b_{ij} = a_{ij}$  for every  $i, j$ . Subtracting  $a_{ij}$  from both sides of this *real-number* equation gives  $b_{ij} = 0$  for every  $i, j$ . Since every entry of  $B$  is 0,  $B = 0$ , the zero matrix. (Equivalently: add  $-A$  to both sides using Theorem 1.2.5  $-(-A) + (A + B) = (-A) + A \Rightarrow 0 + B = 0 \Rightarrow B = 0$ .)

**Problem 1.2.6 (Challenge)**

Find a nonzero  $2 \times 2$  matrix  $A$  with  $A^2 = 0$  but  $A \neq 0$ .

**Solution**

Take  $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Then

$$A^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0(0) + 1(0) & 0(1) + 1(0) \\ 0(0) + 0(0) & 0(1) + 0(0) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0,$$

yet  $A \neq 0$ . This is exactly the nilpotent phenomenon you'll meet again in §1.3 – no nonzero real number squares to 0, but plenty of nonzero matrices do.

**3 Solutions – §1.3 (Some More Special Matrices)****Problem 1.3.1**

Verify  $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$  is symmetric and  $\begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$  is skew-symmetric, directly from the definitions.

**Solution**

For  $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ : transposing swaps the off-diagonal entries  $a_{12} = 1$  and  $a_{21} = 1$  – but they're already equal, so  $A^t = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = A$ . Since  $A^t = A$ ,  $A$  is **symmetric**.

For  $K = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix}$ :  $K^t = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ . Compare with  $-K = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ : they match, so  $K^t = -K$ , confirming  $K$  is **skew-symmetric**.

**Problem 1.3.2**

Confirm  $A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$  is nilpotent by computing  $A^2$  and  $A^3$  yourself.

**Solution**

Multiplying row by column:

$$A^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (\text{nonzero}), \quad A^3 = A^2 \cdot A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0.$$

Since  $A^2 \neq 0$  but  $A^3 = 0$ ,  $A$  is nilpotent with **order of nilpotency** 3.

### Problem 1.3.3

Find a  $2 \times 2$  idempotent matrix that is diagonal but is not  $I_2$  or 0.

### Solution

For a diagonal matrix  $D = \text{diag}(d_1, d_2)$ ,  $D^2 = \text{diag}(d_1^2, d_2^2)$ , so  $D^2 = D$  forces  $d_i^2 = d_i$  for each  $i$ , i.e.  $d_i(d_i - 1) = 0$ , so each  $d_i$  must be 0 or 1 – there's no other option. Choosing one diagonal entry 0 and the other 1 gives a valid answer that's neither  $I_2$  nor 0:

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad D^2 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = D. \checkmark$$

(The matrix  $\text{diag}(0, 1)$  works equally well.)

### Problem 1.3.4

For  $A = \begin{pmatrix} 4 & 2 \\ 0 & -2 \end{pmatrix}$ , compute  $S = \frac{1}{2}(A + A^t)$  and  $T = \frac{1}{2}(A - A^t)$ , and check  $S + T = A$ .

### Solution

$$A^t = \begin{pmatrix} 4 & 0 \\ 2 & -2 \end{pmatrix}.$$

$$S = \frac{1}{2} \left[ \begin{pmatrix} 4 & 2 \\ 0 & -2 \end{pmatrix} + \begin{pmatrix} 4 & 0 \\ 2 & -2 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 8 & 2 \\ 2 & -4 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 1 & -2 \end{pmatrix},$$

$$T = \frac{1}{2} \left[ \begin{pmatrix} 4 & 2 \\ 0 & -2 \end{pmatrix} - \begin{pmatrix} 4 & 0 \\ 2 & -2 \end{pmatrix} \right] = \frac{1}{2} \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Check:  $S + T = \begin{pmatrix} 4 & 1 \\ 1 & -2 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 0 & -2 \end{pmatrix} = A$ .  $\checkmark$  Also note  $S^t = S$  (symmetric) and  $T^t = -T$  (skew-symmetric), as expected.

### Problem 1.3.5

Split a  $4 \times 4$  matrix of your choice into four  $2 \times 2$  blocks and verify Theorem 1.3.6 against a second matrix of your choice.

### Solution

This problem is open-ended – any valid choice works, as long as the block sizes line up. Here is one complete worked instance you can check your own attempt against. Let

$$A = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

Split  $A = [P \ Q]$  by columns and  $B = \begin{pmatrix} H \\ K \end{pmatrix}$  by rows:

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad Q = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}, \quad K = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

Computing  $PH + QK$  block-by-block and comparing with the direct product  $AB$  (both computed independently) gives the *same* matrix in both cases:

$$AB = PH + QK = \begin{pmatrix} 3 & 3 & 4 & 4 \\ 1 & 1 & 3 & 1 \\ 1 & 5 & 1 & 3 \\ 1 & 2 & 3 & 2 \end{pmatrix}. \checkmark$$

If your own two matrices give matching answers by both routes, you've verified the theorem correctly.

### Problem 1.3.6 (Challenge)

Prove that every diagonal entry of a skew-symmetric matrix must be 0, directly from  $A^t = -A$ .

### Solution

By definition of transpose,  $(A^t)_{ii} = a_{ii}$  for every diagonal position (transposing never moves a diagonal entry). The skew-symmetry condition  $A^t = -A$  says every entry of  $A^t$  equals  $-1$  times the corresponding entry of  $A$ ; applying this at position  $(i, i)$  gives

$$a_{ii} = (A^t)_{ii} = -a_{ii}.$$

Adding  $a_{ii}$  to both sides:  $2a_{ii} = 0$ , so  $a_{ii} = 0$ . Since  $i$  was arbitrary, **every** diagonal entry is 0. ■

## 4 Solutions – §1.4 (Matrices over Complex Numbers)

### Problem 1.4.1

For  $A = \begin{pmatrix} 2-i & 4 \\ 0 & 3+i \end{pmatrix}$ , find  $\overline{A}$  and  $A^*$ .

### Solution

Conjugating every entry:

$$\overline{A} = \begin{pmatrix} 2+i & 4 \\ 0 & 3-i \end{pmatrix}.$$

For  $A^*$ , transpose  $\overline{A}$  (equivalently, transpose  $A$  then conjugate):

$$A^* = \begin{pmatrix} 2+i & 0 \\ 4 & 3-i \end{pmatrix}.$$

### Problem 1.4.2

Is  $\begin{pmatrix} 5 & 2+3i \\ 2-3i & -1 \end{pmatrix}$  Hermitian? Justify using Definition 1.4.1.

### Solution

Let  $H = \begin{pmatrix} 5 & 2+3i \\ 2-3i & -1 \end{pmatrix}$ . Compute  $H^*$ : the  $(1,1)$ -entry becomes  $\overline{5} = 5$ , the  $(2,2)$ -entry becomes  $\overline{-1} = -1$ , and the off-diagonal entries swap position and conjugate:  $(1,2)$ -entry of  $H^*$  is  $\overline{(2,1)\text{-entry of } H} = \overline{2-3i} = 2+3i$ , and the  $(2,1)$ -entry of  $H^*$  is  $\overline{(1,2)\text{-entry of } H} = \overline{2+3i} = 2-3i$ . So

$$H^* = \begin{pmatrix} 5 & 2+3i \\ 2-3i & -1 \end{pmatrix} = H.$$

Since  $H^* = H$ , **yes,  $H$  is Hermitian.** (Notice the pattern: real numbers on the diagonal, and the two off-diagonal entries are complex conjugates of each other – this is always what a Hermitian matrix looks like.)

### Problem 1.4.3

Verify directly that  $I_n$  is unitary for any  $n$ .

### Solution

$I_n$  has only real entries (0s and 1s), so conjugating changes nothing:  $\overline{I_n} = I_n$ . Transposing  $I_n$  also changes nothing (it's symmetric), so  $I_n^* = \overline{I_n}^t = I_n^t = I_n$ . Then

$$I_n^* I_n = I_n I_n = I_n = I_n I_n = I_n I_n^*.$$

Since  $I_n^* I_n = I_n = I_n I_n^*$ , by Definition 1.4.1 (unitary condition)  $I_n$  is **unitary**. ■

### Problem 1.4.4

Using  $A^* = -A$ , prove that every diagonal entry of a skew-Hermitian matrix must be purely imaginary.

### Solution

By definition of conjugate transpose,  $(A^*)_{ii} = \overline{a_{ii}}$ . The skew-Hermitian condition  $A^* = -A$  at position  $(i, i)$  gives

$$\overline{a_{ii}} = -a_{ii}.$$

Write  $a_{ii} = x + yi$  for real  $x, y$ . Then  $\overline{a_{ii}} = x - yi$ , and the equation becomes

$$x - yi = -(x + yi) = -x - yi.$$

Comparing real parts:  $x = -x$ , so  $2x = 0$ , i.e.  $x = 0$ . (The imaginary parts already agree automatically:  $-yi = -yi$ .) So  $a_{ii} = 0 + yi = yi$  – purely imaginary (possibly 0, when  $y = 0$  too). Since  $i$  was arbitrary, this holds for **every** diagonal entry. ■

### Problem 1.4.5 (Challenge)

Show that if  $A$  is unitary, then for any  $u, v \in \mathbb{C}^n$ , multiplying both by  $A$  does not change their inner product  $u^*v$ .

**Solution**

We need to show  $(Au)^*(Av) = u^*v$ . Using the identity  $(Au)^* = u^*A^*$  (conjugate-transpose reverses and conjugate-transposes each factor – the complex analogue of  $(AB)^t = B^tA^t$ ):

$$(Au)^*(Av) = (u^*A^*)(Av) = u^*(A^*A)v.$$

Since  $A$  is unitary,  $A^*A = I$  by definition. So

$$(Au)^*(Av) = u^*Iv = u^*v.$$

This confirms that applying a unitary matrix to two vectors leaves their inner product exactly unchanged – which is precisely why unitary (and, in the real case, orthogonal) matrices are the matrices that preserve lengths and angles. ■

*That covers every practice problem from Chapter 1. Send me Chapter 2 whenever you're ready and I'll build its notes (and, if you'd like, another solutions file) the same way.*